Satyam

satyam23495@iiitd.ac.in | +91-9811230374 | github.com/itzsam-lol | leetcode.com/u/samuraiiii | linkedin.com/in/satyam564774288

Education

Indraprastha Institute of Information Technology Delhi (IIITD) — New Delhi, India
B.Tech in CS-AI — *2023 - 2027*

Mira Model School — New Delhi, India
High School Diploma (CBSE) — *2023*

Technical Skills

Computer Science Fundamentals: Data Structures, Algorithms, Complexity Analysis (Big O), Object-Oriented Design, Operating Systems, Computer Architecture, Computer Networks, Database Systems

Languages: C++, Python, Java, C, TypeScript, JavaScript, SQL, RISC-V Assembly

Distributed Systems & Backend: Service-Oriented Architecture, Multi-Tiered Systems, Microservices, Distributed Job Queues, Fault-Tolerant Design, High-Concurrency Systems, REST APIs, FastAPI, Express, Node.js; **Relational Databases** - PostgreSQL, SQLite, Prisma ORM, Schema Design; Redis (in-memory indexing), Object Storage (S3/MinIO)

Optimization & Machine Learning: Linear Programming (MILP), Google OR-Tools, Graph Algorithms (Dijkstra, A*), Combinatorial Optimization; scikit-learn, XGBoost, Random Forest, Predictive Modelling, Data Mining

Cloud, DevOps & Practices: AWS EC2, Docker Compose, GitHub Actions (CI/CD), Linux/UNIX, Firebase, Vercel; Agile Development, Code Review, Git, **300+ LeetCode** (Graphs, DP, Trees, Heaps)

Experience

IIIT Delhi — New Delhi, India
Teaching Assistant - DSA (CSE102, C++) & Computer Networks (CSE232) — *Jan 2026 - Present*

- Mentor **100+ undergraduates** across two courses; design assignments and rubrics covering **data structures, algorithms, complexity analysis, and object-oriented design**; conduct weekly sessions on graph traversal, dynamic programming, and network protocols - articulating complex technical concepts to varied skill levels.

CoSY Lab, IIIT Delhi — New Delhi, India
Research Intern | Guide: Prof. Ganesh Bagler — *May 2025 - July 2025*

- Engineered **high-throughput data ingestion and query pipelines** for Foodoscope processing millions of structured records at scale; built visualisation dashboards enabling multi-dimensional dataset exploration across active research objectives.
- Benchmarked pipeline outputs against domain ground truths via **statistical validation**, ensuring data integrity and reproducibility; extended the Python codebase with new analytical modules under an **open-ended research brief** with no predefined solution path.

Fam (formerly FamPay) — Remote
Software Testing Intern — *May 2024 - July 2024*

- Executed regression test suites across Android and iOS builds each sprint in an **agile release cycle**; identified edge-case failures in payment, onboarding, and card-management flows via structured defect triage; authored reproducible bug reports with root-cause analysis contributing to pre-release stability.

Projects

SynchroPulse AI - Blr Traffic Prediction & Optimization | *Python, OR-Tools, FastAPI, Docker, AWS EC2* | GitHub

- **Applied Mixed-Integer Linear Programming** (Google OR-Tools) to optimize resource allocation on **real Bengaluru traffic data**, computing optimal deployment plans in **under 10 seconds**; integrated custom **Dijkstra/A* routing** over dynamically updated graphs.
- Trained a **predictive model on real-world traffic data** to forecast congestion **180 minutes ahead**, built on a distributed spatial graph pipeline (OSMnx + NetworkX); **deployed live on AWS EC2** via Docker Compose with a FastAPI service layer.

CodeForge - Fault-Tolerant Distributed Platform | *TypeScript, Express, Next.js 15, PostgreSQL, Redis, Docker*

- **Architected a multi-tiered distributed system** decoupling code execution, job orchestration, and analytics into independent services; **distributed job queues** route concurrent grading loads to isolated execution workers, with **Redis indexing**, **PostgreSQL** durable state, and **object storage** for artifacts.
- Engineered **fault tolerance and security**: implemented recovery mechanisms using delayed-job deadlines and periodic state sweeps; secured untrusted code execution using **isolated sandboxes**, restricted worker pools, and revocable session tokens.

PGIP - Predictive Triage for Gene Perturbation Experiments | *Python, scikit-learn, Random Forest, ESM-2* GitHub

- **ROC-AUC 0.86, Spearman ρ 0.65** (preprint in preparation): designed a two-stage ML pipeline predicting which experiments are worth running *before* incurring their cost - **formulated the problem framing** from a high-level research question, mining multi-modal features across three independent large-scale datasets.

RISC-V Assembler & CPU Simulator | *Python, C++, Computer Architecture, OOP* | GitHub

- Built a complete **RV32I toolchain** - two-pass assembler with label resolution and forward-reference handling, plus a cycle-accurate simulator implementing instruction decoding, **register file and memory management**, and ALU execution. Validated by **147 tests** covering opcodes and boundary conditions.

Udyox AI - Production Backend, 18+ Live Clients | *FastAPI, React 19, Whisper, LLMs, SQLite* | GitHub

- Designed and deployed a **modular service backend** serving **18+ live clients with zero downtime**; low-latency pipeline (speech recognition to LLM intent extraction to confirmation-first database write) achieving **2-3s end-to-end response** under real production load.

Leadership & Achievements

- **Algorithms: 300+ LeetCode** problems - graphs, dynamic programming, trees, heaps, and complexity analysis.
- **Vice Chairperson, IEEE-IIITD Student Branch** (Nov 2024 - Aug 2025): led strategy and operations, managing **50+ volunteers** across technical events, workshops, and industry outreach.
- **Co-Founder, Ignite Room:** scaled a tech community to **10,000+ students**; built and maintains the Ignite Room and CiPD (IIIT Delhi) web platforms; Lead Organizer, Hack The Flame national hackathon (**500+ participants**).
- **Formal Verification:** Proved RTL correctness properties on a SystemVerilog design using SymbiYosys, applying constraint modeling to verify system boundaries.
- **Awards:** Creative Student Award and Academic Scholar Award, Mira Model School; Research Contributor, Foodoscope, CoSY Lab IIIT Delhi under Prof. Ganesh Bagler.